

Task 1: Common Hardware Issues and Symptoms

No.	Hardware Issue	Typical Symptom
1	No Display	The monitor shows a blank screen, and the PC may not complete POST.
2	Faulty RAM	The computer may restart repeatedly or fail to boot properly.
3	Hard Drive Failure	The system may become slow or show errors while accessing files.
4	Power Supply Problem	The PC may suddenly shut down or fail to start.
5	Overheating	The computer may become very hot, slow down, or shut down unexpectedly.

Conclusion

Common hardware problems can cause different symptoms such as no display, slow performance,

unexpected shutdowns, or failure to boot. Identifying the symptoms helps to find the possible faulty component.

That Is Not Charging Troubleshooting a laptop

Step 1- check the charger

First, i would check the charger and cable for any visible damage or loose connection.

Step 2- check the power socket

I would plug the charger into another working power socket to make sure the socket is not the problem.

Step 3- check the charging port

I would inspect the laptop's charging port and make sure it is clean, and the charger fits properly.

step 4- check the battery

If the battery is removable, i would switch off the laptop, remove the battery and check it for any visible damage.

Step 5 – Check Charger Output

Using a multimeter, I would carefully check the charger's output voltage and compare it with the voltage written on the charger label.

Step 6 – Test the Connection

I would reconnect the charger and check whether the laptop shows a charging indicator.

Result:

These basic checks can help identify whether the problem is with the power socket, charger, charging port, or battery.

Testing USB Charger Voltage with a Multimeter

I used a multimeter set to DC voltage mode to check the output voltage of a standard USB phone charger.

Measured Voltage: 5.08V

Expected Range: 4.75V – 5.25V

Result:

The measured voltage of 5.08V is within the expected range of 4.75V to 5.25V. Therefore, the USB charger output is within the normal range.

Conclusion:

The multimeter helps to check whether the

charger is providing the correct voltage to the connected device.

Testing the 12V Power Supply Rail

Step 1 – Disconnect the Power Supply

First, I would switch off the PC and disconnect the power cable. I would disconnect the ATX connector from the motherboard before testing.

Step 2 – Set the Multimeter

I would set the multimeter to DC voltage (V-) mode.

Step 3 – Identify the Wires

The yellow wire is the 12V rail, and the black wire is ground.

Step 4 – Measure the Voltage

I would place the multimeter probes between the yellow and black wires and record the voltage shown on the display.

Measured Voltage: 12.08V

Step 5 – Check the Result

The expected 12V rail should normally be around 12V. A reading of 12.08V is above the 11.5V limit mentioned in the task.

Conclusion

If the 12V rail measured below 11.5V, it could indicate a faulty or weak power supply (SMPS/PSU), which may cause random shutdowns or system instability.